nuclear power india target

11 sites

India aims to achieve **100 GWe (Gigawatts electric) of nuclear power capacity by 2047** as part of its Net Zero 2070 strategy. This goal, supported by the Nuclear Energy Mission (Union Budget 2025–26), seeks to significantly increase the current capacity of ~8.8 GW to enhance energy security. ICRA Ratings +3

Key Nuclear Power Targets & Initiatives:

- **100 GW Target by 2047:** The overarching goal is to achieve 100 GWe of nuclear capacity by the 100th year of independence (2047).
- **Intermediate Target (2031-32):** India's installed nuclear capacity is set to increase to 22,380 MW by 2031–32 from 8,780 MW, with 13.6 GW of capacity currently under construction.
- **Fleet Mode Operation:** To meet targets, India is focusing on building 700 MW indigenous Pressurised Heavy Water Reactors (PHWRs) in "fleet mode" to reduce commissioning times, with 10 reactors currently in various stages.

ent has allocated ₹20,000 crore to
to develop at least five indigenous

- **Private Sector Participation:** The SHANTI (Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India) Bill, 2025, facilitates private sector participation to boost capacity expansion.
- **Repurposing Old Plants:** The government is considering using land from retired coal-fired thermal power plants for setting up nuclear reactors. PIB +8

The current nuclear power share is about 2% of India's total installed capacity, which the ambitious 2047 target aims to increase substantially, requiring a massive scaling up of fuel supply and project implementation. ICRA Ratings

1Gw costs ?

12 sites

The capital cost for **1 GW (1,000 MW)** of nuclear power capacity in India is estimated **between ₹14,200 crore and ₹25,000 crore** ($1.7 billion to $3 billion), depending on the technology and project scale. orfonline.org +1

While India builds some of the most cost-effective reactors globally, pricing varies significantly between indigenous and foreign-assisted projects:

Capital Cost Estimates (per GW)

- **Indigenous PHWRs (Pressurised Heavy Water Reactors):**
  - The Central Electricity Authority (CEA) projected costs to rise to **₹14.2 crore per MW** (approx. **₹14,200 crore per GW**) by 2026–27.
  - Other estimates for these 700 MW indigenous units range higher, between **₹16 crore and ₹25 crore per MW** (up to **₹25,000 crore per GW**).
- **Foreign-Assisted Projects (e.g., Kudankulam):**
  - Costs for these larger 1,000 MW units have increased over time due to liability insurance and technology changes.
  - **Units 5 & 6** are budgeted at approximately **₹49,621 crore** for 2 GW, or **₹24,810 crore per GW**. Wikipedia +3

Cost Benchmarks for Specific Projects

- **Fleet Mode (700 MW Units):** India is building 10 indigenous reactors (7 GW total) at a sanctioned cost of **₹1.05 lakh crore**, averaging roughly **₹15,000 crore per GW**.
- **International Comparison:** India's construction cost for indigenous reactors is roughly **$1,700 per kW** (₹14.8 crore/MW), which is 30–40% lower than the global average of $2,500–$3,000+ per kW. PIB +4

Operational Economy

- **Generation Cost:** Once built, nuclear is highly competitive. Older plants like Tarapur produce power at **₹2 per unit**, while newer ones like Kudankulam range from **₹4 to ₹6 per unit**.
- **Long Life:** These costs are spread over an exceptionally long operational life of **60 to 100 years**. orfonline.org +1

Would you like to know more about the **private sector's role** in funding these projects through the new Atomic Energy Bill 2025?

major components by revenue percentage

9 sites

In nuclear power projects, the **overnight capital cost** (construction and equipment) accounts for the vast majority of total project value. Once operational, the levelised cost of energy (LCOE) is dominated by these initial investment costs rather than ongoing operations. Wikipedia +2

The major components of a nuclear power plant's capital cost are typically broken down as follows:

1. Capital Cost Breakdown (Construction Phase)

Based on industry standards for large-scale reactors, the initial investment is distributed across these primary areas:

Component Category	Estimated % of Capital Cost	Key Elements Included
Onsite Labour	**25%**	Installation, assembly, and construction workforce.
Mechanical Equipment	**16%**	Pumps, valves, piping, and heat exchangers.
Nuclear Steam	**12%**	The reactor core, pressure vessels, and steam

Supply (NSSS)		generators.
Electrical & Generating	**12%**	Turbines, generators, and power distribution systems.
Construction Materials	**12%**	High-grade steel, specialized concrete, and containment structures.
Project Management	**10%**	Engineering, design, and site supervision.
Instrumentation & Control	**8%**	Software, monitoring systems, and control room tech.
Other (Fuel/Services)	**5%**	Initial fuel load (approx. 3%) and miscellaneous services.

2. Lifecycle Revenue/Expenditure Share

When looking at the total cost to produce 1 kWh of electricity over the plant's **60+ year life**, the "revenue" (or cost recovered) is split very differently than in coal or gas plants: Wikipedia +1

- **Investment/Capital Recovery (60–80%):** Because nuclear plants have very high upfront costs and take 5–8 years to build, most of the revenue generated goes toward paying back the initial construction loans and interest.
- **Operating & Maintenance (15–25%):** Covers the high-skilled workforce required for 24/7 safe operations.
- **Fuel Costs (10–15%):** Nuclear fuel is a relatively small portion of the cost compared to fossil fuels (where fuel can be 70–80% of the cost).
- **Decommissioning (2%):** A small fraction is typically set aside from every unit sold to fund the eventual dismantling of the plant. World Nuclear Asso... +3

3. Strategic Shifts in India

To optimize these percentages, India is shifting toward:

- **Fleet Mode:** Building identical 700 MW reactors in batches to reduce **labour and project management** costs through "learning by doing".
- **Small Modular Reactors (SMRs):** These aim to shift more of the "site labour" (25%)

into **factory manufacturing**, which is typically cheaper and faster. hlk.in +2

Would you like to see a comparison of these costs against **solar or thermal power** projects in India?

AI responses may include mistakes. For financial advice, consult a professional. Learn more

what are the companies benefitted in each segment?

57 sites

India's nuclear expansion to **100 GW by 2047** involves a specialized supply chain where specific companies dominate based on their engineering and manufacturing capabilities. The key beneficiaries across major segments include:

1. Nuclear Steam Supply System (NSSS) & Core Components

This segment includes the reactor core, pressure vessels, and critical "nuclear island" equipment.

- Larsen & Toubro (L&T): The dominant player with over **50% market share** in steam generators and end shields. It is the first Indian company with ASME N-Stamp certification for nuclear-grade manufacturing.
- Walchandnagar Industries: A legacy partner of the Department of Atomic Energy (DAE). It manufactures critical core equipment like **calandrias, end shields, and dump tanks** for both PHWRs and Fast Breeder Reactors.
- **MTAR Technologies**: A niche specialist in high-precision engineering for the reactor core, including fuel machining heads and coolant assemblies. Equitymaster +55

2. Turbine Island & Electrical Systems

This covers the secondary side of the plant, converting steam into electricity.

- Bharat Heavy Electricals Ltd (BHEL): The primary supplier of **turbine generator sets**, having equipped nearly **50% of India's current nuclear capacity**. It recently secured a

major ₹108 billion order for six turbine islands.

- **Power Mech Projects**: Historically focused on thermal power, it entered the nuclear space with a contract for civil works on the **turbine island package** at the Kaiga project. Equitymaster +2

3. Engineering & Civil Construction

Civil works are critical for housing the reactor and providing safety containment.

- **Hindustan Construction Company (HCC)**: India's leading nuclear constructor, responsible for building facilities that account for **over 60% of the nation's nuclear capacity**.
- Megha Engineering & Infrastructure (MEIL): Recently awarded a major EPC contract for the construction of Kaiga Units 5 and 6. World Nuclear Asso... +2

4. Supporting Systems (Pumps, Valves, & Forgings)

- **KSB Ltd**: The only approved supplier for **primary coolant pumps** in nuclear reactors in India.
- **Patels Airtemp**: One of only three Indian firms with **N-NPT authorisation** for manufacturing nuclear-grade heat exchangers and pressure vessels.
- Venus Pipes & Tubes: Growing supplier of high-grade stainless steel pipes and tubes for reactor cooling systems. Torus Digital +1

5. Future Operators & Private Partners

Following the SHANTI Act (2025), private players are now moving toward plant ownership and operation. Ventura Securities

- **NTPC Ltd**: Formed the **ASHVINI joint venture** with NPCIL to build the Mahi Banswara project.
- Tata Power: Actively exploring the **Small Modular Reactor (SMR)** sector and seeking government approval for private involvement. World Nuclear News +3

Would you like to explore how the recent **SHANTI Bill** changes the **investment rules** for these private companies?

AI responses may include mistakes. For financial advice, consult a professional. Learn more

yes

12 sites

The SHANTI (Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India) Bill, 2025 overhauls India's nuclear sector by repealing the Atomic Energy Act, 1962, and the Civil Liability for Nuclear Damage Act, 2010. This move ends the long-standing state monopoly of the **Nuclear Power Corporation of India Ltd (NPCIL)**, allowing private companies to own and operate nuclear plants for the first time. PRSIndia +2

Revised Investment & Ownership Rules

- **Private Participation:** Private Indian companies and joint ventures can now build, own, operate, and decommission nuclear power plants.
- **Equity Caps:** Private investment in nuclear power projects is subject to a **49% cap** to ensure majority strategic control remains with government-linked entities.
- **Foreign Investment:** The Bill allows for up to **49% Foreign Direct Investment (FDI)** in specified nuclear activities, potentially attracting global technology partners like Westinghouse or GE.
- **Specialised Activities:** Private players can now participate in non-strategic fuel-cycle activities, including **fuel fabrication** and the exploration of atomic minerals. PRSIndia +7

Liability Reform (Unlocking Bankability)

- **Removal of Supplier Liability:** A major change is the elimination of the "right of recourse" against equipment suppliers for patent or latent defects. This was a long-standing deterrent for international vendors who feared unlimited liability.
- **Graded Liability Caps:** Instead of a flat cap, the Bill introduces tiered liability based on reactor thermal power.

- **Large Plants (>3600 MW):** Capped at ₹3,000 crore.
- **Medium Plants (1500–3600 MW):** Capped at ₹1,500 crore.
- **SMRs (<150 MW):** Capped at ₹100 crore.

- **Insurance Mandate:** Only private operators are required to maintain insurance or financial security to cover these liability caps. Digital Sansad +4

Regulatory & Strategic Safeguards

- **Statutory Regulator:** The **Atomic Energy Regulatory Board (AERB)** has been granted statutory status, making it independent from the Department of Atomic Energy (DAE) and accountable to Parliament.
- **Reserved Functions:** The central government retains exclusive control over "strategic" functions, including **uranium enrichment**, heavy water production, and high-level radioactive waste management.
- **Dispute Resolution:** A dedicated **Atomic Energy Redressal Advisory Council** and a specialised tribunal have been proposed to handle sector-specific disputes, bypassing traditional civil court delays. PRSIndia +3

Would you like to see a list of **SMR (Small Modular Reactor) projects** currently being fast-tracked under this new bill?

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The Big Picture: India Opens Nuclear Sector to Private...

The Asia Group

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